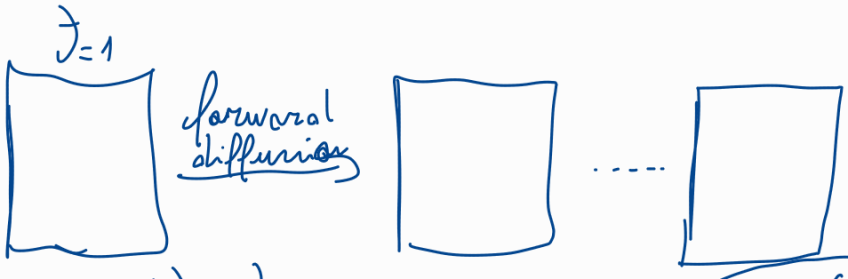
FROM YOUTUBE

DIFFUSION MODELS

Learn how to reverse noise (get an image of pure noise as input and denoise it step-by-step into a real image)

TRAINING

- FORWARD DIFFUSION



$x_1 \sim$ dataset

$$x_t = a_t x_1 + b_t \varepsilon \quad \varepsilon \sim N(0,1)$$

Keep the image values of x_t in the same range as x_1 we use a_t → consider only a fraction of x_1

generate a noise version of x_1 (called x_t)

add gaussian noise ε to each pixel of x_1

NOISE SCHEDULE → amount of noise that we add depends on a predefined SCHEDULE (usually INCREASE when t INCREASE)

b_1 → NO NOISE

b_t → PARTIALLY CORRUPTED NOISE

b_0 → ALL NOISE

- BACKWARD DIFFUSION (with NN → U-NET / → TRANSFORMER)

Let the NN predict the noise ε that we added in the FORWARD DIFFUSION

↳ LOSS $\mathcal{L} = \| \hat{\varepsilon} - \varepsilon \|_2^2$

$\hat{\varepsilon}$: predicted noise

ε: true noise that we actually added

INFERENCE

we use ε predicted by NN to reverse the diffusion process

Takes as input a complete noise image and let the model to predict the noise present in the image. Then subtract the predicted noise to the image to get a cleaner image.

⇓

We don't subtract all the noise at once because the NN learns how to denoise a specific amount of noise at time-step t ⇒ we denoise in small-steps (not just one)

⇓

It learns what noise should I remove to slightly move this sample towards the data distribution

- NEED hundreds of denoising steps
- EXTREMELY STABLE TO TRAIN (loss simple and well behaved)

FLOW MATCHING

→ skip the randomness of diffusion and replace with smooth deterministic flow

⇓

LEARN a deterministic flow (smooth path) that directly transports points from the noise distribution to the data distribution

STARTING POINT

→ VELOCITY FIELD

$$x_1 = x_0 + \int_0^1 v(x_t, t)\, dt$$

x_1 ↓ CLEAN IMAGE

↓

train a neural network that learns the velocity field v → we can use it to move from x_0 to x_1

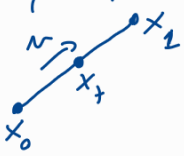
TRAINING

(CLEAN DATA) x_1

(NOISE) $x_0 \sim N(0,1)$

$$x_t = (1-t)x_0 + t\,x_1 \quad \text{(INTERPOLATION)}$$

→ compute the velocity field (GROUND TRUTH)



↓

$$\text{MSE}(\hat{v}, v)$$

↑

x_t → NN predicts $\hat{v}$

↓

it learns to PREDICT the correct FLOW FIELD from noise to data

- We need just 5-15 forward passes during inference (less than DIFFUSION model)

↓

FLOW MATCHING can generate images in a fraction of time